Virtualization

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Agenda

- Introduction to virtualization
- Hypervisor
- Types of virtualization
- Working of a virtual machine
- Setting up a virtual machine
- Benefits of virtualization
- Disadvantages of virtualization

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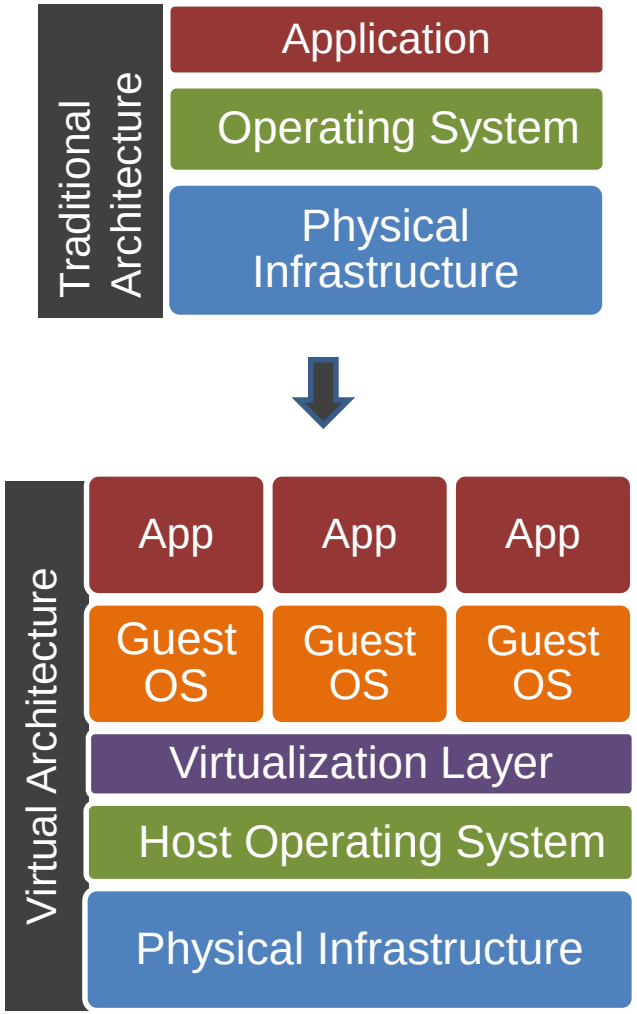
Introduction

- Virtualization is the process of running a virtual instance of a computer system in a layer abstracted from the actual hardware.
- Multiple operating systems can be run on the same computer simultaneously.
- A virtual machine is the emulation of a computer system.
- Uses Cases:
  - Run applications
  - Segmenting large servers/systems
  - Effective Resource utilization



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Need for Virtualization

Virtualization

- Segmenting large systems
- Run multiple applications
- Dividing a large server
- The cloud advantage
- Administrative costs
- Resource utilization

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Hypervisor

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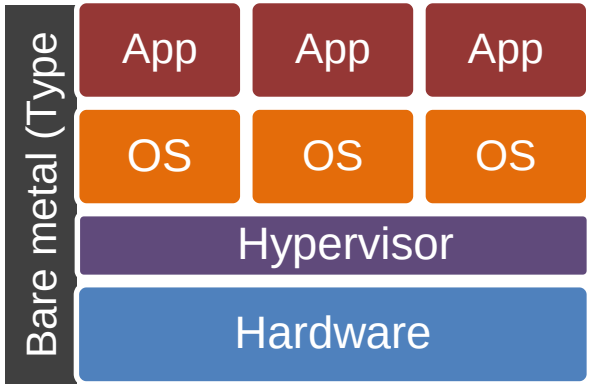
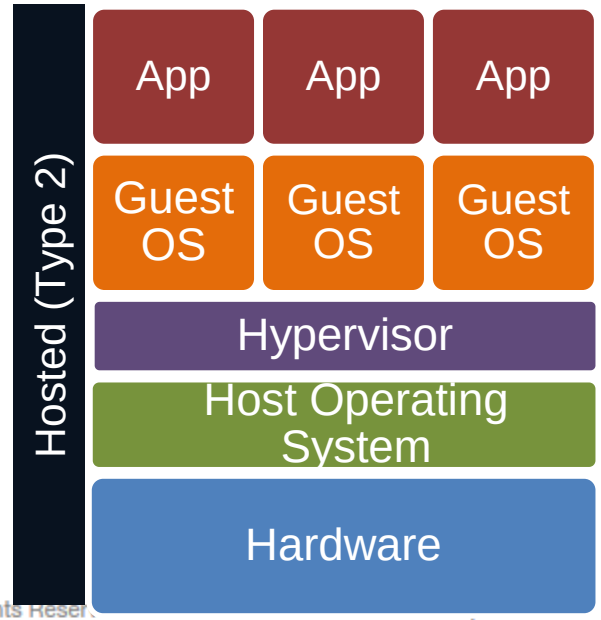
- Implementing a virtual machine (VM) on a host computer requires a hypervisor.
- Hypervisor is a program which helps to run and create a virtual machine.
- There two types of hypervisor:
  - Bare metal hypervisor (Type 1)- These hypervisors utilize the hardware of the host to emulate a system (VM) on top of it.
  - Hosted hypervisor (Type 2)- The VM running via a hosted hypervisor runs like an application.





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Popular Hypervisors

- VMware ESXi/vSphere
- Microsoft Hyper-V
- Citrix XenServer
- Oracle VirtualBox
- Parallels Hypervisor
- Red Hat Enterprise Virtualization

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Types of Virtualization

- Desktop Virtualization
- Network Virtualization
- Server Virtualization
- Storage Virtualization
- Application Virtualization
- Data Virtualization

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Working of a virtual machine

Step 1
- Detach required resources

Step 2
- Segment available resources

Step 3
- Computations

Step 4
- Additional Resources

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Setting up a virtual machine

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- Prerequisites: System requirements (minimum)- Intel I3 or above, RAM-4GB or more.
- For 64 bit OS the hardware virtualization support needs to be enabled in BIOS.
  - Enter BIOS
  - Click on Advanced
  - Click CPU Configuration
  - Turn on Intel VMX virtualization technology
  - Download Kali Linux/Ubuntu as necessary (select 64 bit/32 bit as necessary).
- Download and install Oracle Virtual box.
- Download and install the same Virtual box extended platform.
- Run the installed virtual box.
- Create a virtual machine by adding name, machine folder, type and version (kali/ubuntu).

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Setting up a virtual machine

- Select amount of RAM (Do not allocate a lot of RAM.
- Select “create a virtual hard disk now”.
- Select Dynamically allocated for storage on hard drive.
- Select file location for storage.
- Click on Start virtual machine. Select the iso of Kali Linux iso and click start.
- Perform a graphical install for Kali Linux.
- Complete the setup of Kali Linux.
- Name your Kali Linux system, set-up its username and password.
- Select default desktop environment and select default options during the installation.
- Once done, Kali Linux will start-up on the VM.

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Benefits of Virtualization

- Increased efficiency of available resources
- Simpler management of software policies
- Reduction in downtime
- Faster provisioning of applications
- Increased responsiveness
- Saves operational cost
- Enhances data and information security
- Effective recovery from disaster

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Disadvantages of Virtualization

- The efficiency when compared to real machines is low.
- One can observe performance concerns if the host system is not powerful enough for the requisite task.
- Specifically for cloud computing, a virtual machine would have higher operating costs
- In case the host system is not well protected, the virtual machine can be easily infected
- If a fault has occurred, it might be difficult to determine the exact reason for the same.

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Summary

- Virtualization and its types
- Use of Hypervisor
- Setting up a virtual machine
- Benefits and disadvantages of virtualization

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